

MSSQL Stored Procedure Assignment

Scenario

A telecom company manages subscriber registration, package updates, recharge transactions, and reporting through a SQL Server database.

The DBA wants to use **Stored Procedures** to standardize database operations, improve security, reduce repeated SQL code, and support reporting.

Question 1 — Insert New Subscriber

Create a stored procedure to insert a new subscriber with:

- Subscriber name
- Phone number
- Package name
- Balance

```
30 CREATE PROCEDURE sp_InsertSubscriber
31     @name VARCHAR(100),
32     @phone VARCHAR(20),
33     @package VARCHAR(50),
34     @balance MONEY
35 AS
36 BEGIN
37     INSERT INTO subscribers (subscriber_name, phone_number, package_name, balance)
38     VALUES (@name, @phone, @package, @balance);
39 END;
40 GO
41
42 -- Test Call
43 EXEC sp_InsertSubscriber 'Mya Mya', '094444444', 'Premium', 8000;
44 GO
```

Adds a new subscriber record with name, phone, package, and balance.

- Ensures standardized data entry.
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Question 2 — View All Subscribers

Create a stored procedure to display all subscribers.

```
49 CREATE PROCEDURE sp_ViewSubscribers
50 AS
51 BEGIN
52     SELECT * FROM subscribers;
53 END;
54 GO
55
56 -- Test Call
57 EXEC sp_ViewSubscribers;
58 GO
59
```

100 % 1 0

Results Messages

	subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
1	1	Aung Aung	091111111	Basic	5000.00	2026-06-03 20:05:21.383
2	2	Su Su	092222222	Premium	10000.00	2026-06-03 20:05:21.383
3	3	Kyaw Kyaw	093333333	Student	3000.00	2026-06-03 20:05:21.383
4	4	Mya Mya	094444444	Premium	8000.00	2026-06-03 20:05:31.790

Displays all subscriber records.

- Useful for quick reporting.

Question 3 — Search Subscriber by Phone Number

Create a stored procedure that accepts a phone number and returns subscriber details.

```
63 CREATE PROCEDURE sp_SearchSubscriberByPhone
64     @phone VARCHAR(20)
65 AS
66 BEGIN
67     SELECT * FROM subscribers WHERE phone_number = @phone;
68 END;
69 GO
70
71 -- Test Call
72 EXEC sp_SearchSubscriberByPhone '09222222';
73 GO
74
```

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Results Messages

	subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
1	2	Su Su	092222222	Premium	10000.00	2026-06-03 20:05:21.383

Finds subscriber details using phone number.

- Supports customer service lookups.

Question 4 — Update Subscriber Package

Create a stored procedure to update a subscriber's package by subscriber ID.

```
78 CREATE PROCEDURE sp_UpdateSubscriberPackage
79     @id INT,
80     @package VARCHAR(50)
81 AS
82 BEGIN
83     UPDATE subscribers SET package_name = @package WHERE subscriber_id = @id;
84 END;
85 GO
86
87 -- Test Call
88 EXEC sp_UpdateSubscriberPackage 1, 'Gold';
89 GO
```

Changes a subscriber's package by ID.

- Useful for upgrades/downgrades.

Question 5 — Recharge Subscriber Balance

Create a stored procedure that adds recharge amount to subscriber balance.

```
94 CREATE PROCEDURE sp_RechargeSubscriber
95     @id INT,
96     @amount MONEY
97 AS
98 BEGIN
99     UPDATE subscribers SET balance = balance + @amount WHERE subscriber_id = @id;
100 END;
101 GO
102
103 -- Test Call
104 EXEC sp_RechargeSubscriber 2, 2000;
105 GO
```

Adds recharge amount to balance.

- Simulates top-up transactions.
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Question 6 — Delete Subscriber

Create a stored procedure to delete a subscriber by subscriber ID.

```
110 CREATE PROCEDURE sp_DeleteSubscriber
111     @id INT
112 AS
113 BEGIN
114     DELETE FROM subscribers WHERE subscriber_id = @id;
115 END;
116 GO
117
118 -- Test Call
119 EXEC sp_DeleteSubscriber 3;
120 GO
```

Removes a subscriber record by ID.

- Used for deactivation.

Question 7 — Package-Based Report

Create a stored procedure to show subscribers by package name.

```
125 CREATE PROCEDURE sp_ReportByPackage
126     @package VARCHAR(50)
127 AS
128 BEGIN
129     SELECT * FROM subscribers WHERE package_name = @package;
130 END;
131 GO
132
133 -- Test Call
134 EXEC sp_ReportByPackage 'Premium';
135 GO
136
```

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Results

Messages

	subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
1	2	Su Su	092222222	Premium	12000.00	2026-06-03 20:05:21.383
2	4	Mya Mya	094444444	Premium	8000.00	2026-06-03 20:05:31.790

Lists subscribers filtered by package name.

- Supports package-based reporting.
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Question 8 — Balance Report

Create a stored procedure to show subscribers whose balance is greater than a given amount.

```
140 CREATE PROCEDURE sp_ReportByBalance
141     @amount MONEY
142 AS
143 BEGIN
144     SELECT * FROM subscribers WHERE balance > @amount;
145 END;
146 GO
147
148 -- Test Call
149 EXEC sp_ReportByBalance 5000;
150 GO
```

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Results Messages

	subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
1	2	Su Su	092222222	Premium	12000.00	2026-06-03 20:05:21.383
2	4	Mya Mya	094444444	Premium	8000.00	2026-06-03 20:05:31.790

Shows subscribers with balance above a threshold.

- Useful for financial analysis.

Database and Table Setup

```
CREATE DATABASE TelecomSPDB;
GO
```

```
USE TelecomSPDB;
GO
```

```
CREATE TABLE subscribers (
    subscriber_id INT PRIMARY KEY IDENTITY(1,1),
    subscriber_name VARCHAR(100) NOT NULL,
    phone_number VARCHAR(20) UNIQUE NOT NULL,
    package_name VARCHAR(50),
    balance MONEY DEFAULT 0,
    created_at DATETIME DEFAULT GETDATE()
);
GO
```

```
INSERT INTO subscribers
(subscriber_name, phone_number, package_name, balance)
```

```
VALUES
('Aung Aung', '091111111', 'Basic', 5000),
('Su Su', '092222222', 'Premium', 10000),
('Kyaw Kyaw', '093333333', 'Student', 3000);
GO
```

Advanced Stored Procedure Questions

Question 9 — Count Subscribers by Package

```
155 CREATE PROCEDURE sp_CountSubscribersByPackage
156 AS
157 BEGIN
158     SELECT package_name, COUNT(*) AS SubscriberCount
159     FROM subscribers
160     GROUP BY package_name;
161 END;
162 GO
163
164 -- Test Call
165 EXEC sp_CountSubscribersByPackage;
166 GO
167
```

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Results Messages

	package_name	SubscriberCount
1	Gold	1
2	Premium	2

Aggregates subscriber counts per package.

- Helps measure package popularity.
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Question 10 — Validate Recharge Amount

Create a stored procedure that prevents recharge amount less than 1000 MMK.

```
171 CREATE PROCEDURE sp_ValidateRecharge
172     @id INT,
173     @amount MONEY
174 AS
175 BEGIN
176     IF @amount < 1000
177         PRINT 'Recharge amount must be at least 1000 MMK';
178     ELSE
179         UPDATE subscribers SET balance = balance + @amount WHERE subscriber_id = @id;
180 END;
181 GO
182
183 -- Test Calls
184 EXEC sp_ValidateRecharge 1, 500; -- Should print warning
```

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Messages

Recharge amount must be at least 1000 MMK

```
185 EXEC sp_ValidateRecharge 1, 1500; -- Should update balance
186 GO
```

Prevents recharge amounts below 1000 MMK.

- Ensures business rules are enforced.
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